

CS100 Fall 2023

Quiz 1

Oct 26, 2023

Answer the questions according to the C17 standard.

1. (15 points) Your name: _____. Your student ID: _____.
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2. (15 points) Select the piece(s) of code that have undefined behaviors.

- A. `int random(void) { int x; return x; }`
`int main(void) { printf("The random number is %d\n", random()); }`
- B. `int table[10];`
`int exists_in_table(int value) {`
 `for (int i = 0; i <= 10; ++i)`
 `if (table[i] == value) return 1;`
 `return 0;`
`}`
- C. `double *ptr = NULL; free(ptr);`
- D. `void func(int, int);`
`int main(void) { int x = 42; func(x, x + 1); }`
- E. `void print_as_int(double x) {`
 `printf("%d\n", x);`
`}`
- F. `void foo(int *ptr) { // ptr may be null`
 `if (*ptr == 42 && ptr) do_something();`
`}`

3. (10 points) Design a function `swap` that can swap two pointers (both of type `int *`).

```
int i = 10, j = 15, *p1 = &i, *p2 = &j;
swap(/* you design this */);
```

After this call to `swap`, `p1` should be pointing to `j` and `p2` should be pointing to `i`.

- A. `void swap(int *p1, int *p2) { int *tmp = p1; p1 = p2; p2 = tmp; }`
To call this function: `swap(p1, p2);`
- B. `void swap(int **p1, int **p2) { int *tmp = *p1; *p1 = *p2; *p2 = tmp; }`
To call this function: `swap(&p1, &p2);`
- C. `void swap(int **p1, int **p2) { int **tmp = p1; p1 = p2; p2 = tmp; }`
To call this function: `swap(&p1, &p2);`
- D. `void swap(int *p1, int *p2) { int **tmp = &p1; &p1 = &p2; &p2 = tmp; }`
To call this function: `swap(&*p1, &*p2);`

4. (10 points) Suppose `n` is an unsigned integer. Select the piece(s) of code where `foo` is always called exactly `n` times.

A. `while (n--) foo();`
B. `for (unsigned i = 0; i <= n; ++i, foo()) {}`
C. `unsigned i = 0;`
`do { foo(); ++i; } while (i < n);`
D. `unsigned i = 0;`
`while (1) {`
`if (++i >= n) break;`
`foo();`
`}`

5. (10 points) Read the following code. Select the correct statement(s).

```
#include <stdio.h>
int x = 42; // (1)
int incre(int x) { // (2)
    return x + 1;
}
int main(void) {
    int x; // (3)
    x = 15;
    if (x == 15) {
        int x = incre(10);
        printf("%d\n", x++); // (4)
    }
    printf("%d\n", incre(x)); // (5)
    return 0;
}
```

- A. The `x` in (1) and (2) refer to the same variable.
B. In (3), the value of `x` is 42 because it was initialized to 42 in (1).
C. The value printed by (4) is 11.
D. The value printed by (5) is 16.
E. The value printed by (5) is 13.
6. (20 points) For each piece of code, write down the type of `var`.

(a) `double* a, b[10], var;`

(a) double

(b) `void foo(int var[3][4]) { /* ... */ }`

(b) int (*)[4]

(c) `int var[3][4], *x = &var[0][0];`

(c) int [3][4]

(d) `char *var[] = {"hello", "world"};`

(d) char *[2]